

```

Name: Rahul KT
Roll No: 2024BCS0224
Date: 12-SEP-2025
*/
/* Drop tables if they already exist (to avoid errors when re-running) */
DROP TABLE IF EXISTS MovieTheaters;
DROP TABLE IF EXISTS Movies;

/* Create Movies table */
CREATE TABLE Movies (
    Code INT PRIMARY KEY,
    Title VARCHAR(100),
    Rating VARCHAR(10)
);
/* Create MovieTheaters table */
CREATE TABLE MovieTheaters (
    Code INT PRIMARY KEY,
    Name VARCHAR(100),
    Movie INT,
    FOREIGN KEY (Movie) REFERENCES Movies(Code)
);
/* Insert into Movies */
INSERT INTO Movies (Code, Title, Rating) VALUES
(1, 'Citizen Kane', 'PG'),
(2, 'Singin' in the Rain', 'G'),
(3, 'The Wizard of Oz', 'G'),
(4, 'The Quiet Man', NULL),
(5, 'North by Northwest', 'NR'),
(6, 'The Last Tango in Paris', 'NC-17'),
(7, 'Some Like it Hot', 'PG-13'),
(8, 'A Night at the Opera', NULL);
/* Insert into MovieTheaters */
INSERT INTO MovieTheaters (Code, Name, Movie) VALUES
(1, 'Odeon', 5),
(2, 'Imperial', 1),
(3, 'Majestic', NULL),
(4, 'Royale', 6),
(5, 'Paraiso', 3),
(6, 'Nickelodeon', NULL);
/* Verify data */
SELECT * FROM MovieTheaters;
SELECT * FROM Movies;

```

RAHUL KT
2024BCS0224

Output:

+-----+		
Code	Name	Movie
+-----+		
1	Odeon	5
2	Imperial	1
3	Majestic	NULL
4	Royale	6
5	Paraiso	3
6	Nickelodeon	NULL
+-----+		
+-----+		
Code	Title	Rating
+-----+		
1	Citizen Kane	PG
2	Singin' in the Rain	G
3	The Wizard of Oz	G
4	The Quiet Man	NULL
5	North by Northwest	NR
6	The Last Tango in Paris	NC-17
7	Some Like it Hot	PG-13
8	A Night at the Opera	NULL
+-----+		

```

1  /* Assignment Submission
2  | Name: RAHUL KT
3  | Roll No: 2024BCS0224
4  | Date: 12-SEP-2025
5  */
6
7  /* -----
8  | Step 1: Drop existing tables
9  | ----- */
10 DROP TABLE IF EXISTS MovieTheaters;
11 DROP TABLE IF EXISTS Movies;
12
13 /* -----
14 | Step 2: Create Tables
15 | ----- */
16 CREATE TABLE Movies (
17     Code INT PRIMARY KEY,
18     Title VARCHAR(255) NOT NULL,
19     Rating VARCHAR(255)
20 );
21
22 CREATE TABLE MovieTheaters (
23     Code INT PRIMARY KEY,
24     Name VARCHAR(255) NOT NULL,
25     Movie INT,
26     FOREIGN KEY (Movie) REFERENCES Movies(Code)
27 );
28
29 /* -----
30 | Step 3: Insert Sample Data
31 | (Movies first, then Theaters)
32 | ----- */
33
34 /* Insert into Movies */
35 INSERT INTO Movies (Code, Title, Rating) VALUES
36 (1, 'Citizen Kane', 'PG'),
37 (2, 'Singin' in the Rain', 'G'),
38 (3, 'The Wizard of Oz', 'G'),
39 (4, 'The Quiet Man', NULL),
40 (5, 'North by Northwest', 'NR'),
41 (6, 'The Last Tango in Paris', 'NC-17'),
42 (7, 'Some Like it Hot', 'PG-13'),

```

STDIN

Input for the program (Optional)

Output:

```

+-----+
| Title |
+-----+
| Citizen Kane |
| Singin' in the Rain |
| The Wizard of Oz |
| The Quiet Man |
| North by Northwest |
| The Last Tango in Paris |
| Some Like it Hot |
| A Night at the Opera |
+-----+
+-----+
| Rating |
+-----+
| PG |
| G |
| NULL |
| NR |
| NC-17 |
| PG-13 |
+-----+
+-----+
| Title |
+-----+
| The Quiet Man |
| A Night at the Opera |
+-----+

```

```

39 (4, 'The Quiet Man', NULL),
40 (5, 'North by Northwest', 'NR'),
41 (6, 'The Last Tango in Paris', 'NC-17'),
42 (7, 'Some Like it Hot', 'PG-13'),
43 (8, 'A Night at the Opera', NULL);
44
45 /* Insert into MovieTheaters */
46 INSERT INTO MovieTheaters (Code, Name, Movie) VALUES
47 (1, 'Odeon', 5),
48 (2, 'Imperial', 1),
49 (3, 'Majestic', NULL),
50 (4, 'Royale', 6),
51 (5, 'Paraiso', 3),
52 (6, 'Nickelodeon', NULL);
53
54 /* -----
55 | Step 4: Assignment Queries
56 -----*/
57
58 /* 1. Select the title of all movies */
59 SELECT Title FROM Movies;
60
61 /* 2. Show all the distinct ratings in the database */
62 SELECT DISTINCT Rating FROM Movies;
63
64 /* 3. Show all unrated movies */
65 SELECT Title FROM Movies WHERE Rating IS NULL;
66
67 /* 4. Select all movie theaters that are not currently showing a movie */
68 SELECT Name FROM MovieTheaters WHERE Movie IS NULL;
69
70 /* 5. Select all data from all movie theaters and, additionally,
71 | the data from the movie that is being shown in the theater */
72 SELECT mt.Code, mt.Name, m.Title, m.Rating
73 FROM MovieTheaters mt
74 LEFT JOIN Movies m ON mt.Movie = m.Code;
75
76 /* 6. Select all data from all movies and, if that movie is being shown
77 | in a theater, show the data from the theater */
78 SELECT m.Code, m.Title, m.Rating, mt.Name AS Theater
79 FROM Movies m
80 LEFT JOIN MovieTheaters mt ON m.Code = mt.Movie;

```

STDIN

Input for the program (Optional)

```

+-----+
| Majestic |
| Nickelodeon |
+-----+

+-----+-----+-----+-----+
| Code | Name      | Title                | Rating |
+-----+-----+-----+-----+
| 1 | Odeon     | North by Northwest   | NR     |
| 2 | Imperial  | Citizen Kane         | PG     |
| 3 | Majestic  | NULL                 | NULL   |
| 4 | Royale    | The Last Tango in Paris | NC-17  |
| 5 | Paraiso   | The Wizard of Oz     | G      |
| 6 | Nickelodeon | NULL                 | NULL   |
+-----+-----+-----+-----+

+-----+-----+-----+-----+
| Code | Title                | Rating | Theater |
+-----+-----+-----+-----+
| 1 | Citizen Kane         | PG     | Imperial |
| 2 | Singin' in the Rain  | G      | NULL     |
| 3 | The Wizard of Oz     | G      | Paraiso  |
| 4 | The Quiet Man        | NULL   | NULL     |
| 5 | North by Northwest   | NR     | Odeon    |
| 6 | The Last Tango in Paris | NC-17  | Royale   |
| 7 | Some Like it Hot     | PG-13  | NULL     |
| 8 | A Night at the Opera | NULL   | NULL     |
+-----+-----+-----+-----+

+-----+
| Title                |
+-----+
| Singin' in the Rain  |

```

```

FROM Movies m
LEFT JOIN MovieTheaters mt ON m.Code = mt.Movie;

/* 7. Show the titles of movies not currently being shown in any theaters */
SELECT Title
FROM Movies m
LEFT JOIN MovieTheaters mt ON m.Code = mt.Movie
WHERE mt.Movie IS NULL;

/* 8. Add the unrated movie "One, Two, Three" */
INSERT INTO Movies (Code, Title, Rating) VALUES
(9, 'One, Two, Three', NULL);

/* 9. Set the rating of all unrated movies to "G" */
UPDATE Movies
SET Rating = 'G'
WHERE Rating IS NULL;

/* 10. Remove movie theaters projecting movies rated "NC-17" */
DELETE mt
FROM MovieTheaters mt
JOIN Movies m ON mt.Movie = m.Code
WHERE m.Rating = 'NC-17';

```

Code	Title	Rating	Theater
1	Citizen Kane	PG	Imperial
2	Singin' in the Rain	G	NULL
3	The Wizard of Oz	G	Paraiso
4	The Quiet Man	NULL	NULL
5	North by Northwest	NR	Odeon
6	The Last Tango in Paris	NC-17	Royale
7	Some Like it Hot	PG-13	NULL
8	A Night at the Opera	NULL	NULL
Title			
Singin' in the Rain			
The Quiet Man			
Some Like it Hot			
A Night at the Opera			